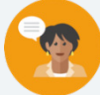


6.6 Similarity in Transformations

Lesson Introduction

 Benchmarks	MA.912.GR.2.8 Apply an appropriate transformation to map one figure onto another to justify that the two figures are similar. MA.912.GR.2.3 Identify a sequence of transformations that will map a given figure onto itself or onto another congruent or similar figure.		 Learning Targets	- I can map one similar figure onto another and justify that the figures are similar. - I can describe a sequence of transformations that will map the preimage onto the image.
 Benchmark Coherence	Building On N/A		Working Toward N/A	
 Essential Questions	- What is true about the corresponding angles of similar figures? - What is true about the corresponding sides of similar figures? - What criteria must a figure have to be similar to another?			
 Lesson Narrative	This lesson guides students through exploring the definition of similar figures by discovering that similar figures have congruent angles and proportional sides. Students then use this criterion to determine if given images are similar.			
 Component Summary	6.6.1 Warm-Up (~5 min.)	Dilating a polygon on an isometric grid (<i>SE p. 311</i>)	6.6.3 Guided Practice (5-10 min.)	Identifying and justifying whether given figures are similar (<i>SE p. 314</i>)
	6.6.2 Exploration (15-20 min.)	Discovering the qualities of similar figures (<i>SE p. 312</i>)	6.6.4 Your Turn! (5-10 min.)	Justifying why figures are similar (<i>SE p. 316</i>)
	6.6.5 Wrap-Up (~5 min.)	Summarizing the lesson by writing a note to a friend (<i>SE p. 317</i>)		
 Resources	Glossary		Materials	
	<u>Key Terms:</u> Similarity <u>Additional Terms:</u> N/A		- Patty Paper (Optional) Graph Paper (Optional) Rulers (Optional) Cut Outs of the Images and Preimages	
Additional Preparation If using stations to encourage collaboration, print out multiple copies of each question to have them available at each station.				

 Teacher Tips	Common Misconceptions	Things to Consider
	- Students may reflect the preimage in 6.6.2 Exploration question 1c instead of rotating it. - Students may dilate using a center at the origin instead of at point C' in 6.6.4 Your Turn! question 1.	Students may struggle with 6.6.4 Your Turn! question 1a. Consider creating a handout with scaffolded questions to guide students in determining the position of the final image.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Anticipate, Monitor, Select, Sequence, Connect Throughout the lesson, students may write sequences of transformations that are not commonly used by the majority of the class. While monitoring student work, make note of any different sequences, and select those students to share with the whole class. This purposeful selecting of students will result in students staying engaged, because the selections were different, and not repetitive of what they have already seen or done.	MA.K12.MTR.5.1: Patterns and Structure In the 6.6.1 Warm-Up, students apply previously learned concepts to a new situation, developing an ability to transfer what they have learned to the world around them. In 6.6.2 Exploration, students relate the previously learned concepts of mapping, lengths of segments, and proportions to connect and discover the criteria needed to state two figures are similar.
 Differentiate Instruction	As students progress through 6.6.2 Exploration, students are given an opportunity to foster collaboration and build a mathematical community. Students can engage with the content and explore each question as teams, pairs, or whole-class collaborations. It may be effective to have stations where students physically go to work on each question throughout the lesson. This way, other students who are working on the same thing can offer tips and advice on how to solve parts they already have mastered.	
Lesson Notes:		

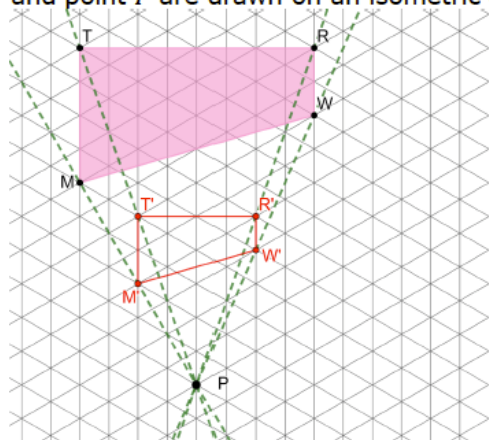
6.6.1 Warm-Up: Dilation on an Isometric Graph

Component Overview: Students apply their skills of drawing the dilation of a figure on a coordinate plane to dilating a figure on an isometric grid.



6.6.1 Warm-Up: Dilation on an Isometric Graph

1. $TRWM$ and point P are drawn on an isometric grid.



- a. Draw a dilation of $TRWM$ centered at point P using a scale factor of $\frac{1}{2}$.

The drawing is completed on the graph in red.

- b. Explain what features on the graph were most useful in helping you determine where the dilation of $TRWM$ should be drawn.

Answers will vary. One thing that helped me was the intersections of the triangles. By counting from point P to each vertex, the new quadrilateral vertex was found by using the midpoints.



This question is a great example of applying something previously learned to a new situation. If time permits, allow students to share their thinking for question 1b. Hearing how others worked through the question can help students make connections that they did not previously see.

6.6.2 Exploration: Similarity

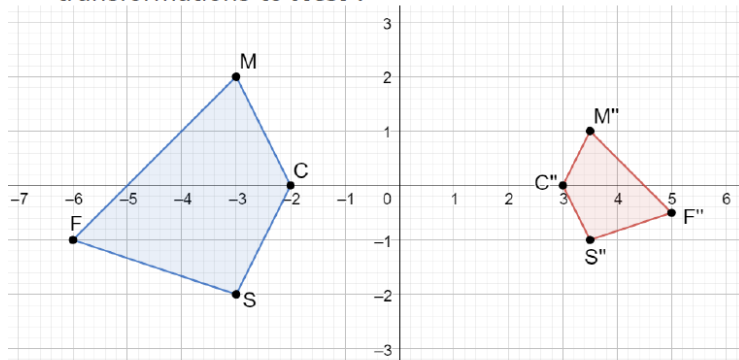
Component Overview: Students explore the corresponding parts of an image that were created from a preimage using a sequence of transformations. Using patty paper, the students determine that corresponding angles are congruent. Using side lengths, the students determine that the corresponding sides have proportional lengths. Connections are made that similar figures are formed when a sequence of transformations includes a dilation and are verified with these two criteria.



6.6.2 Exploration: Similarity

With your partner, complete the following.

1. $M''C''S''F''$ was created by applying a sequence of transformations to $MCSF$.



- a. With patty paper, trace $\angle FMC$ in $MCSF$.
- b. Place the tracing over $\angle F''M''C''$ in $M''C''S''F''$.
- c. What do you notice about $\angle FMC$ and $\angle F''M''C''$?
The angles are congruent.
- d. Repeat the tracing for $\angle C$, $\angle S$, and $\angle F$. What do you notice about each angle and their corresponding angles?
All of the corresponding angles are congruent.
- e. Complete the table.

Angles	Comparison Between Angles
$\angle FMC$ and $\angle F''M''C''$	<i>Congruent</i>
$\angle MCS$ and $\angle M''C''S''$	<i>Congruent</i>
$\angle CSF$ and $\angle C''S''F''$	<i>Congruent</i>
$\angle SFM$ and $\angle S''F''M''$	<i>Congruent</i>



Notice and Wonder

Consider utilizing the instructional routine “Notice and Wonder” prior to students answering question 1. This will allow students to make observations about the figures and understand the context in a low-stakes environment.

6.6.2 Exploration: Similarity (continued)

- f. Determine a sequence of transformations that can be used to map $MCSF$ to $M''C''S''F''$.

Answers may vary. Reflected across the line $x = \frac{1}{2}$ and then dilated by a factor of $\frac{1}{2}$ with point C' as the center of dilation.

- g. Ask two classmates the transformations they determined. Record their transformations and write a summary of their reasoning. *You are the only person who should write in the first two columns of the table.* Have the person initial next to your summary stating your summary was correct. Remind your classmates to include any details on the center of rotation or dilation and any scale factor.

Answers will vary.

Transformations	My Summary of Their Reason	Initials

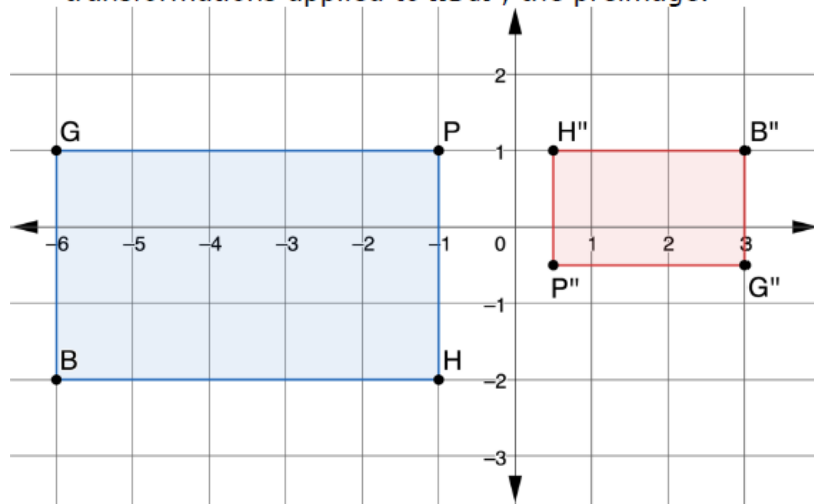
- h. Work with your partner to verify your classmates' transformations are correct.



While monitoring student work, make note of the different sequences that students determined. Have select students share any sequences that were not commonly used by the class to highlight different ideas.

6.6.2 Exploration: Similarity (continued)

2. $H''B''G''P''$ was created using a sequence of transformations applied to $HBGP$, the preimage.



- a. Complete the table.

Preimage	Side Length	Image	Side Length	Ratio of Preimage to Image Lengths
$\overline{GP}$	5 units	$\overline{G''P''}$	2.5 units	$\frac{2}{1}$
$\overline{PH}$	3 units	$\overline{P''H''}$	1.5 units	$\frac{2}{1}$
$\overline{HB}$	5 units	$\overline{H''B''}$	2.5 units	$\frac{2}{1}$
$\overline{BG}$	3 units	$\overline{B''G''}$	1.5 units	$\frac{2}{1}$



Students may choose to write the ratio as 2 to 1, $\frac{2}{1}$, or as 2:1. All are acceptable. Continue to use different representations to keep students familiar with all forms.

- b. Describe the relationship between the ratios of the preimage to the image.

All of the ratios of a side length to the corresponding side length are the same, 2:1.

6.6.2 Exploration: Similarity (continued)

- c. Determine the sequence of transformations that was used to create $H''B''G''P''$ from $HBGP$.

The figure was rotated 180° about the origin and then dilated a scale factor of $\frac{1}{2}$ with the center of dilation at the origin.

- d. Discuss with your partner how to describe how $MCSF$ is related to $M''C''S''F''$ and how $HBGP$ is related to $H''B''G''P''$. Summarize your discussion.
Answers will vary. In both sets of quadrilaterals, the corresponding angles in the preimage and image are congruent but the corresponding sides are not congruent.

3. What qualities of similar figures do these quadrilaterals have?

They have corresponding angles that are congruent and corresponding sides that are proportional.



Similarity is having exactly the same shape but not necessarily the same size. Two figures are similar if and only if one figure can be obtained from the other by a dilation or a sequence of transformations that includes a dilation.

4. Complete the statements using words from the bank.

Word Bank

congruent	dilation	proportional	similar
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When a sequence of transformations includes a **dilation**, the resulting image of the sequence of transformations is **similar** to the preimage. All of the angles of the resulting image are **congruent** to the corresponding angles of the preimage. The corresponding sides of the image to the preimage are **proportional**.



Students may incorrectly reflect the preimage instead of rotating it since this will produce the same shape. Probe students about the position of the corresponding vertices and their alignment to the image if a reflection is performed.



The key word “similarity” will be essential throughout the rest of Geometry. Make sure students take time to learn this definition.



Consider discussing how the ratio of the corresponding lengths relates to the scale factor of the dilation after students have completed 6.6.2 Exploration.

6.6.3 Guided Practice: Similarity

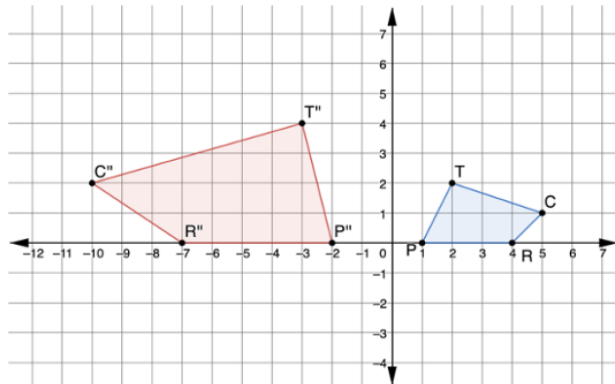
Component Overview: Students observe three preimages and images to determine similarity. If they are not similar, a justification is required.



6.6.3 Guided Practice: Similarity

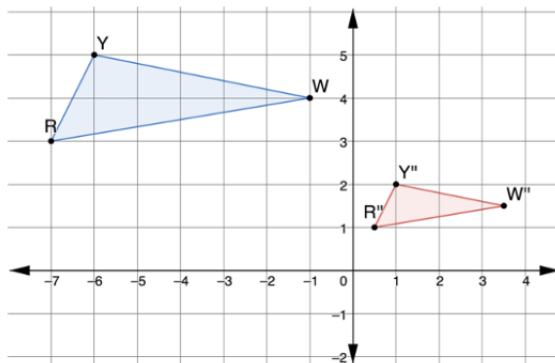
For each pair of figures, write the sequence of transformations that would show the figures are similar. If the figures are not similar, explain why.

1.



The figures are not similar. The ratio of PQ to $P''Q''$, $3:5$, is not equal to the ratio of PR to $P''R''$, $\sqrt{5}:\sqrt{17}$.

2.



These figures are similar. Answers will vary. The preimage has been dilated by a scale factor of $\frac{1}{2}$ with the center of dilation at the origin and then translated right 4 and down 0.5.



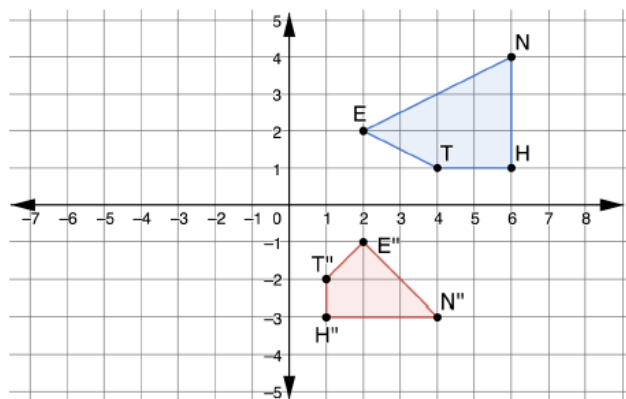
To add a challenge, have students determine the least number of points that can be moved to make any figures that are not similar become similar figures. (Sample answer for question 1: two moves. Move R'' to $(-8,0)$ and move T'' to $(-4,4)$.)



For question 2, students may choose a different sequence of transformations. While monitoring, make note of different methods and highlight those during the review of this question.

6.6.3 Guided Practice: Similarity (continued)

3.



The figures are not similar. The ratio of HT to $H''T''$ is $2:1$, but the ratio of HN to $H''N''$ is $1:1$.



For students who may be struggling, consider providing students with the image after the first transformations. This can help students see the two transformations that were performed to help them determine if the shapes are similar.

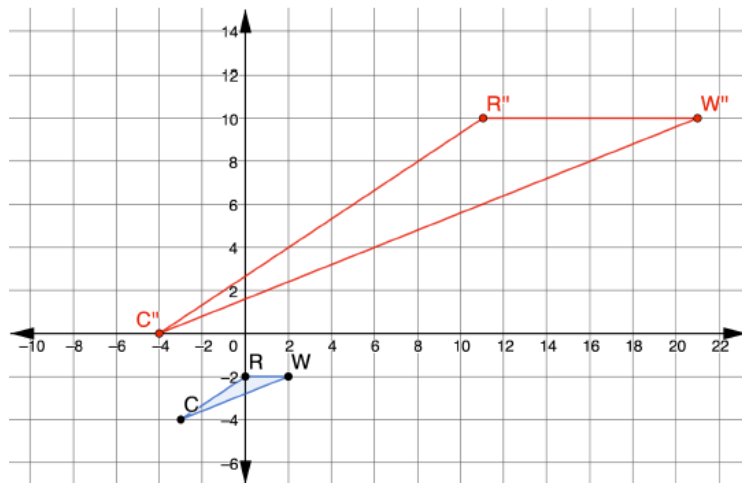
6.6.4 Your Turn!

Component Overview: Students justify whether two figures are similar by two different methods, one by a sequence of transformations and the other by proportionality of sides.



6.6.4 Your Turn!

1. $\triangle RWC$ is shown on the coordinate plane.



- a. Draw $\triangle R''W''C''$, which is a result of a translation up 4 units and left 1 unit, then dilated using a scale factor of 5 centered at C' .
Image is drawn on graph.

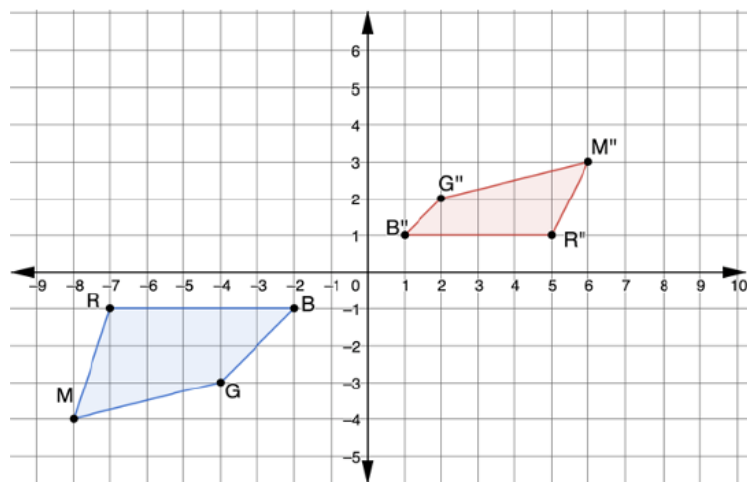
- b. Explain how $\triangle RWC$ is similar to $\triangle R''W''C''$.
The triangles are similar because the final image is the result of a translation, which is a rigid transformation, and a dilation, which preserves angle measure and the corresponding sides from preimage to image have the same ratio.



Students may mistakenly dilate the image centered at the origin instead of point C' . If this happens, have the student re-read the directions aloud to determine their error.

6.6.4 Your Turn! (continued)

2. Carter and Isabella are asked to determine if the quadrilaterals are similar. Their responses are recorded in the table.



Carter's Response	Isabella's Response
Yes, the quadrilaterals are similar because all the corresponding angles are congruent.	No, the quadrilaterals are not similar because the corresponding sides are not proportional.

Is Carter's or Isabella's response correct? Explain your decision.

Answers will vary. Isabella's response is correct. The corresponding sides are not proportional. The ratio of BR to $B'R'$ is $5:4$ but the ratio of RM to $R'M''$ is $\sqrt{10}:\sqrt{5}$.



This would be a great opportunity to develop students' abilities to appropriately communicate with their peers and offer help without providing answers. Consider having the students write a note to Carter that includes one positive statement and one constructive statement.

6.6.5 Wrap-Up: Cool Down

Component Overview: Students summarize the day by writing a note to a friend explaining the lesson.



6.6.5 Wrap-Up: Cool Down

1. Your friend is absent today and will need the notes from class. Write a quick note to help them understand how to determine if a sequence of transformations results in similar figures.

Answers will vary.



Consider offering students the opportunity to create a picture that summarizes the day's lesson. This may be a comic book or a puzzle.